

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. A ball of ice melts so that its surface area decreases at a rate of $2 \text{ cm}^2 / \text{min}$. Find the rate at which the diameter is decreasing when the diameter is 12 cm .

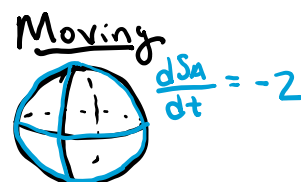
The surface area of a sphere is given by $A = 4\pi r^2$.

We need to relate the surface area to the diameter and so we get $A = 4\pi \left(\frac{d}{2}\right)^2$.

We calculate that $\frac{dA}{dt} = 8\pi \left(\frac{d}{2}\right) \left(\frac{1}{2} \frac{dd}{dt}\right) = 2\pi d \frac{dd}{dt}$

We then evaluate the derivative at the moment when the diameter is 12 cm .

$$(-2) = 2\pi(12) \frac{dd}{dt} \rightarrow \frac{dd}{dt} = -\frac{1}{12\pi} \text{ cm/min}$$



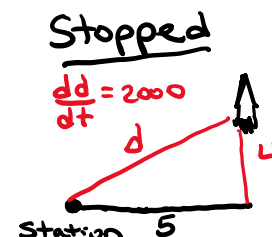
2. A rocket, rising vertically, is tracked by a radar station that is on the ground 5 miles from the launch pad. How fast is the rocket rising when it is 4 mile high and its distance from the radar station is increasing at a rate of 2000 miles/hour?

We can relate the variables in the figure with $h^2 + 5^2 = d^2$.

We calculate $\frac{d}{dt}[h^2 + 5^2] = \frac{d}{dt}[d^2] \rightarrow 2h \frac{dh}{dt} = 2d \frac{dd}{dt}$.

We evaluate this derivative when $h = 4$, $\frac{dd}{dt} = 2000$, & $d = \sqrt{25 + 16} = \sqrt{41}$

$$2(4) \frac{dh}{dt} = 2(\sqrt{41})(2000) \rightarrow \frac{dh}{dt} = \frac{4000\sqrt{41}}{8} = 500\sqrt{41} \text{ miles/hour}$$



3. Wheat is poured through a chute at a rate of $10 \text{ ft}^3 / \text{min}$ and falls in a conical pile whose bottom radius is always half the altitude.

- a) How fast will the radius of the base be increasing when the pile is 8ft high?

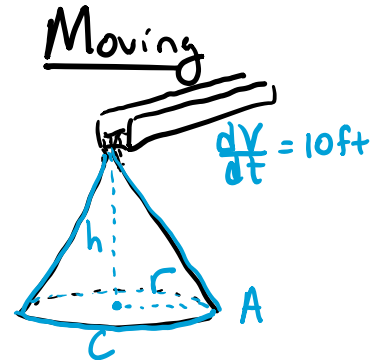
We know that $V = \frac{1}{3} \pi r^2 h$.

We are told that $h = 2r$ and therefore $V = \frac{1}{3} \pi r^2 (2r) = \frac{2\pi}{3} r^3$

We then calculate that $\frac{d}{dt}[V] = \frac{d}{dt}\left[\frac{2\pi}{3} r^3\right] \rightarrow \frac{dV}{dt} = 2\pi r^2 \frac{dr}{dt}$.

We then evaluate this derivative when $\frac{dV}{dt} = 10$ & $r = \frac{8}{2} = 4$.

$$(10) = 2\pi(4)^2 \frac{dr}{dt} \rightarrow \frac{dr}{dt} = \frac{10}{32\pi} = \frac{5}{16\pi} \text{ ft/min}$$



- b) How fast will the circumference of the base be increasing when the pile is 8ft high?

We already know from part (a) that $V = \frac{2\pi}{3} r^3$.

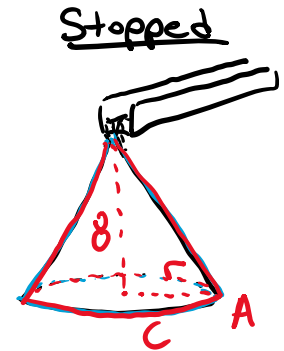
We also know that $C = 2\pi r \rightarrow r = \frac{C}{2\pi}$.

This means that we can write $V = \frac{2\pi}{3} \left(\frac{C}{2\pi}\right)^3 = \frac{1}{12\pi^2} C^3$

We then calculate that $\frac{d}{dt}[V] = \frac{d}{dt}\left[\frac{1}{12\pi^2} C^3\right] \rightarrow \frac{dV}{dt} = \frac{1}{4\pi^2} C^2 \frac{dC}{dt}$.

We then evaluate this derivative when $\frac{dV}{dt} = 10$ & $r = \frac{8}{2} = 4 \rightarrow C = 8\pi$.

$$(10) = \frac{1}{4\pi^2} (8\pi)^2 \frac{dC}{dt} \rightarrow \frac{dC}{dt} = \frac{10}{16} = \frac{5}{8} \text{ ft/min}$$



- c) How fast will the area of the base be increasing when the pile is 8ft high?

We already know from part (a) that $V = \frac{2\pi}{3} r^3$.

We also know that $A = \pi r^2 \rightarrow r = \sqrt{\frac{A}{\pi}}$.

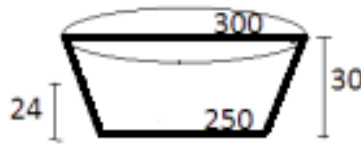
This means that we can write $V = \frac{2\pi}{3} \left(\sqrt{\frac{A}{\pi}}\right)^3 = \frac{2}{3\sqrt{\pi}} A^{3/2}$

We then calculate that $\frac{d}{dt}[V] = \frac{d}{dt}\left[\frac{2}{3\sqrt{\pi}} A^{3/2}\right] \rightarrow \frac{dV}{dt} = \frac{1}{\sqrt{\pi}} A^{1/2} \frac{dA}{dt}$.

We then evaluate this derivative when $\frac{dV}{dt} = 10$ & $r = \frac{8}{2} = 4 \rightarrow A = 16\pi$.

$$(10) = \frac{1}{\sqrt{\pi}} \sqrt{16\pi} \frac{dA}{dt} \rightarrow \frac{dA}{dt} = \frac{10}{4} = \frac{5}{2} \text{ ft}^2/\text{min}$$

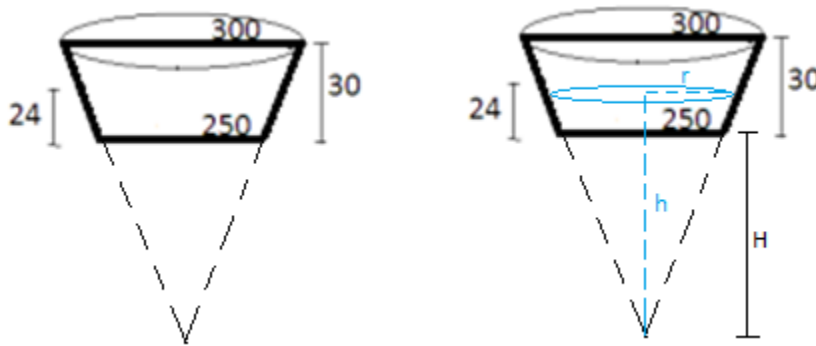
4. A 30ft deep retention pond is roughly shaped like a section of an inverted right circular cone, the bottom having a radius of 250ft and the top having a radius of 300ft (see the diagram below). During a heavy rainstorm, the pond has overcome its discharge tube and has risen to 24ft deep. At that instant the water level is still rising at the rate of about 2inches/hour. If the rain doesn't let up, when would you predict the pond to overflow its banks?



We first need to find a formula for the volume of the water in the tank.

We can find this volume by considering the full conic shape of the water (assuming the tank was extended to a full cone shape), finding its volume, and then subtracting off the portion of this cone that is not a part of this water.

We draw the full conic shape by extending the sides of the tank downward. We then label the heights of the small and large cone as shown in the following diagram.



We note right away, that from similar triangles $\frac{H}{H+30} = \frac{250}{300} \rightarrow 6H = 5H + 150 \rightarrow H = 150$.

This means that H is a constant value.

We also note from similar triangles that $\frac{r}{300} = \frac{h}{H+30} \rightarrow r = \frac{300}{180}h \rightarrow r = \frac{5}{3}h$

So, we derive each of the following volumes.

We see that the volume of the full conic shape (determined by the water) is given by $V = \frac{1}{3}\pi(r)^2 h$.

We see that the volume of the smaller extra conic shape is given by $V = \frac{1}{3}\pi(250)^2 H = 3125000\pi$

Therefore, the volume of the water in the tank is given by

$$V = \frac{1}{3}\pi r^2 h - 3125000\pi = \frac{1}{3}\pi\left(\frac{5}{3}h\right)^2 h - 3125000\pi = \frac{25}{27}\pi h^3 - 3125000\pi.$$

Now we calculate that $\frac{dV}{dt} = \frac{25}{9} \pi h^2 \frac{dh}{dt}$. We find how quickly the volume is changing when the height of the water in the tank is 24ft deep. We see that $h = H + 24 = 174$ and $\frac{dh}{dt} = 2 \text{ in} / \text{hr} = \frac{1}{6} \text{ ft} / \text{hr}$ and so,

$$\frac{dV}{dt} = \frac{25}{9} \pi (174)^2 \left(\frac{1}{6} \right) = \frac{42050}{3} \pi$$

The volume of the rest of the tank is given by:

$$V = \frac{1}{3} \pi (300)^2 (180) - \frac{1}{3} \pi (290)^2 (174) = 522200\pi$$

Therefore, the time it would take to for the pond to overflow is given by

$$\frac{\text{volume of empty portion pond}}{\text{rate at which it is filling up}} = \frac{522200\pi}{\left(\frac{42050}{3} \pi \right)} = \frac{31332}{841} \approx 37.25565$$

So, it will take roughly 37.25565 hours for the pond to overflow.

5. A camera is mounted at a point 100m away from a rocket launching pad. The rocket rises vertically at 28,000 km/hr.
- a) The angle of elevation of the of the camera needs to change at just the right rate to keep it in sight. At what rate does the camera need to rotate upwards (in radians per second) when the rocket is 1 km off the ground?

$$\text{We see that } \tan(\theta) = \frac{h}{100}$$

$$\text{We calculate that } \sec^2(\theta) \frac{d\theta}{dt} = \frac{1}{100} \frac{dh}{dt}.$$

We know that when the rocket is 1 km off the ground

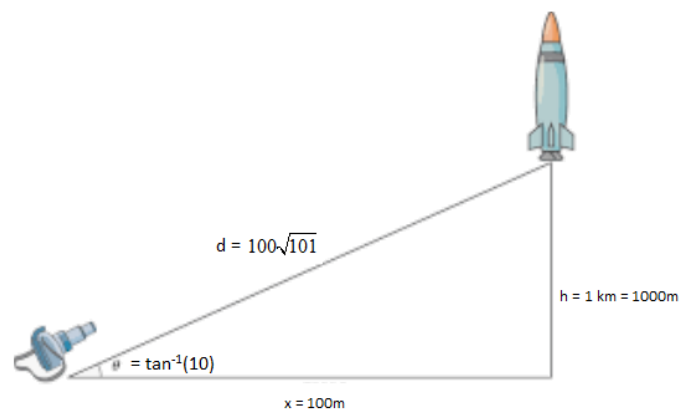
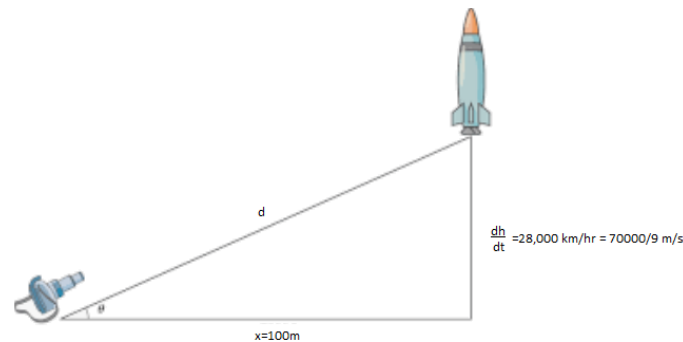
$$\text{that } \tan(\theta) = \frac{1000}{100} \rightarrow \theta = \tan^{-1}(10) \approx 1.4711$$

We know that $\sec(\tan^{-1}(10)) = \sqrt{101}$ and that

$$\frac{28000 \text{ km}}{1 \text{ hr}} \cdot \frac{1 \text{ hr}}{60 \text{ min}} \cdot \frac{1 \text{ min}}{60 \text{ sec}} \cdot \frac{1000 \text{ m}}{1 \text{ km}} = \frac{70000 \text{ m}}{9 \text{ sec}}$$

So,

$$\begin{aligned} \sec^2(\theta) \frac{d\theta}{dt} &= \frac{1}{100} \frac{dh}{dt} \rightarrow (\sqrt{101})^2 \frac{d\theta}{dt} = \frac{1}{100} \frac{70000}{9} \\ &\rightarrow \frac{d\theta}{dt} = \frac{700}{909} \approx 0.7701 \text{ rad} / \text{sec} \end{aligned}$$



b) In addition, to the angle of the camera needing to change, the camera-to-rocket distance is changing constantly, which means the focusing mechanism will also have to change at just the right rate to keep the picture sharp. A lens is able to focus according to the formula $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$, where f represents the focal length of the camera lens, d_o represents the distance to the object we are trying to get in focus, and d_i represents the distance from the lens to the image (loosely speaking how far the lens has to move for the object to remain in focus).

- i. At what rate is the distance between the rocket and camera increasing at the moment in time when the rocket is 1 km off the ground? Express your answer in m/s .

We can relate the sides of the triangle with the equation $100^2 + h^2 = d^2$.

$$\text{Then we have } 2h \frac{dh}{dt} = 2d \frac{dd}{dt}$$

We the rocket is 1 km off the ground we know that $h = 1$ & $d = 100\sqrt{101}$ (see above pictures).

Therefore, we calculate that

$$2h \frac{dh}{dt} = 2d \frac{dd}{dt} \rightarrow 2(1) \left(\frac{70000}{9} \right) = 2(100)\sqrt{101} \frac{dd}{dt} \rightarrow \frac{dd}{dt} = \frac{700}{9\sqrt{101}} m/s \approx 7.7392 m/s$$

- ii. Suppose that the camera lens being used has a focal length of 800mm. At what rate does the lens need to be moving/adjusting at the moment when the rocket is 1 km off the ground?

We can relate the variables we need with the equation $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \rightarrow \frac{1}{0.8} = \frac{1}{d_o} + \frac{1}{d_i}$

$$\text{We calculate that } 0 = -\frac{1}{d_o^2} \frac{dd_o}{dt} - \frac{1}{d_i^2} \frac{dd_i}{dt}.$$

We know, from part (a), that when the rocket is 1 km off the ground

- the distance between the camera and the rocket is $100\sqrt{101}$
- this distance between the camera and the rocket is increasing at a rate of $\frac{dd_o}{dt} = \frac{700}{9\sqrt{101}} m/s$
- The value of d_i can be solved for because we need to have $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$ be true at this instant in time.

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \rightarrow \frac{1}{0.8} = \frac{1}{100\sqrt{101}} + \frac{1}{d_i} \rightarrow d_i = \frac{1}{\frac{1}{100\sqrt{101}} - \frac{5}{4}} \rightarrow d_i = \frac{1-125\sqrt{101}}{100\sqrt{101}}$$

Therefore, we see that

$$\begin{aligned}
0 &= -\frac{1}{d_o^2} \frac{dd_o}{dt} - \frac{1}{d_i^2} \frac{dd_i}{dt} \rightarrow 0 = -\frac{1}{(100\sqrt{101})^2} \left(\frac{700}{9\sqrt{101}} \right) - \frac{1}{\left(\frac{1-125\sqrt{101}}{100\sqrt{101}} \right)^2} \frac{dd_i}{dt} \\
&\rightarrow 0 = -\frac{700}{100^2 \cdot 101 \cdot 9\sqrt{101}} - \left(\frac{100\sqrt{101}}{1-125\sqrt{101}} \right)^2 \frac{dd_i}{dt} \\
&\rightarrow 0 = -\frac{700}{100^2 \cdot 101 \cdot 9\sqrt{101}} - \frac{100^2 \cdot 101}{1-250\sqrt{101}+101} \frac{dd_i}{dt} \\
&\rightarrow \frac{dd_i}{dt} = -\frac{700}{100^2 \cdot 101 \cdot 9\sqrt{101}} \cdot \frac{102-250\sqrt{101}}{100^2 \cdot 101} \\
&\rightarrow \frac{dd_i}{dt} = -\frac{700(102-250\sqrt{101})}{100^4 \cdot 101^2 \cdot 9\sqrt{101}} \\
&\rightarrow \frac{dd_i}{dt} = -\frac{7(102-250\sqrt{101})}{100^3 \cdot 101^2 \cdot 9\sqrt{101}} \approx -0.000000018287 m/s = -0.000018287 mm/s
\end{aligned}$$